

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Thesis title

NGUYỄN DOÃN A

nguyenvanabc@sis.hust.edu.vn

Program: Global ICT

Supervisor: Associate Professor Lê Tùng ABC

Department: Computer Engineering

School: School of Information and Communications Technology

HANOI, 06/2022

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ACKNOWLEDGMENT

[TODO] I would like to express my sincere gratitude to my supervisor, [**Advisor name**], for the continuous guidance throughout this project . . .

Hanoi, [month]/[year]

Student

Duong Phuong Thao

ABSTRACT

[TODO] Write the abstract here as a single coherent paragraph ...

Student

Duong Phuong Thao

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LIST OF ABBREVIATIONS

Abbreviation	Full Expression
API	Giao diện lập trình ứng dụng (Application Programming Interface)
EUD	Phát triển ứng dụng người dùng cuối(End-User Development)
GWT	Công cụ lập trình Javascript bằng Java của Google (Google Web Toolkit)
HTML	Ngôn ngữ đánh dấu siêu văn bản (HyperText Markup Language)
IaaS	Dịch vụ hạ tầng

CHAPTER 1. INTRODUCTION

This chapter introduces the context and motivation of the project, defines its objectives and scope, outlines the tentative solution, and describes the organization of the thesis.

1.1 Motivation

[TODO] Describe the practical context and the problem it raises.

1.2 Objectives and Scope of the Graduation Thesis

[TODO] State objectives, scope and the main functions of the system.

1.3 Tentative Solution

[TODO] Outline the proposed solution direction in a few sentences.

1.4 Thesis Organization

Chapter 2 surveys existing distribution systems and analyses the functional and non-functional requirements. Chapter 3 presents the theoretical background and technologies adopted. Chapter 4 describes the architecture, detailed design, implementation and evaluation of the system. Chapter 5 discusses the main contributions, in particular the FEFO inventory-allocation mechanism. Chapter 6 concludes the thesis and outlines future work.

CHAPTER 2. REQUIREMENT SURVEY AND ANALYSIS

This chapter analyses the current situation, surveys comparable products, and specifies the functional and non-functional requirements of the system.

2.1 Status Survey

[TODO] Survey and compare existing products; justify the required features.

Table 2.1: Comparison of existing distribution-management products

Criterion	Odoo	KiotViet	Sapo	This thesis
Lot/expiry (FEFO) tracking	...	...	...	Yes
Role-based approval workflow	...	...	...	Yes (9 roles)
Delivery planning & shipper app	...	...	...	Yes
Self-hosted / customisable	...	...	...	Yes

2.2 Functional Overview

2.2.1 General Use Case Diagram

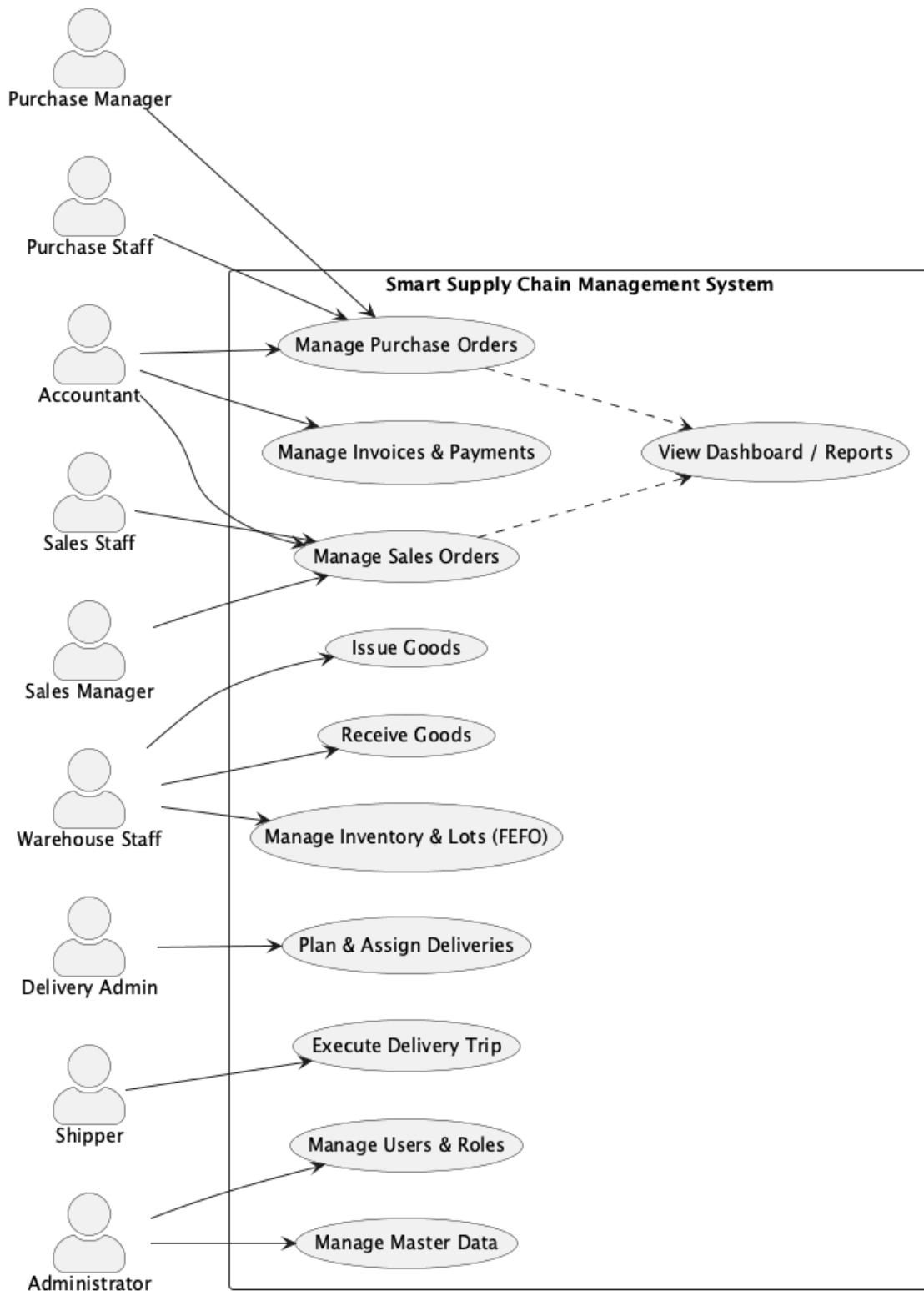


Figure 2.1: General use case diagram of the system

[TODO] Explain the actors (Purchase/Sales Staff & Manager, Warehouse Staff, Delivery Admin, Shipper, Accountant, Administrator) and the main use cases.

2.2.2 Detailed Use Case Diagrams

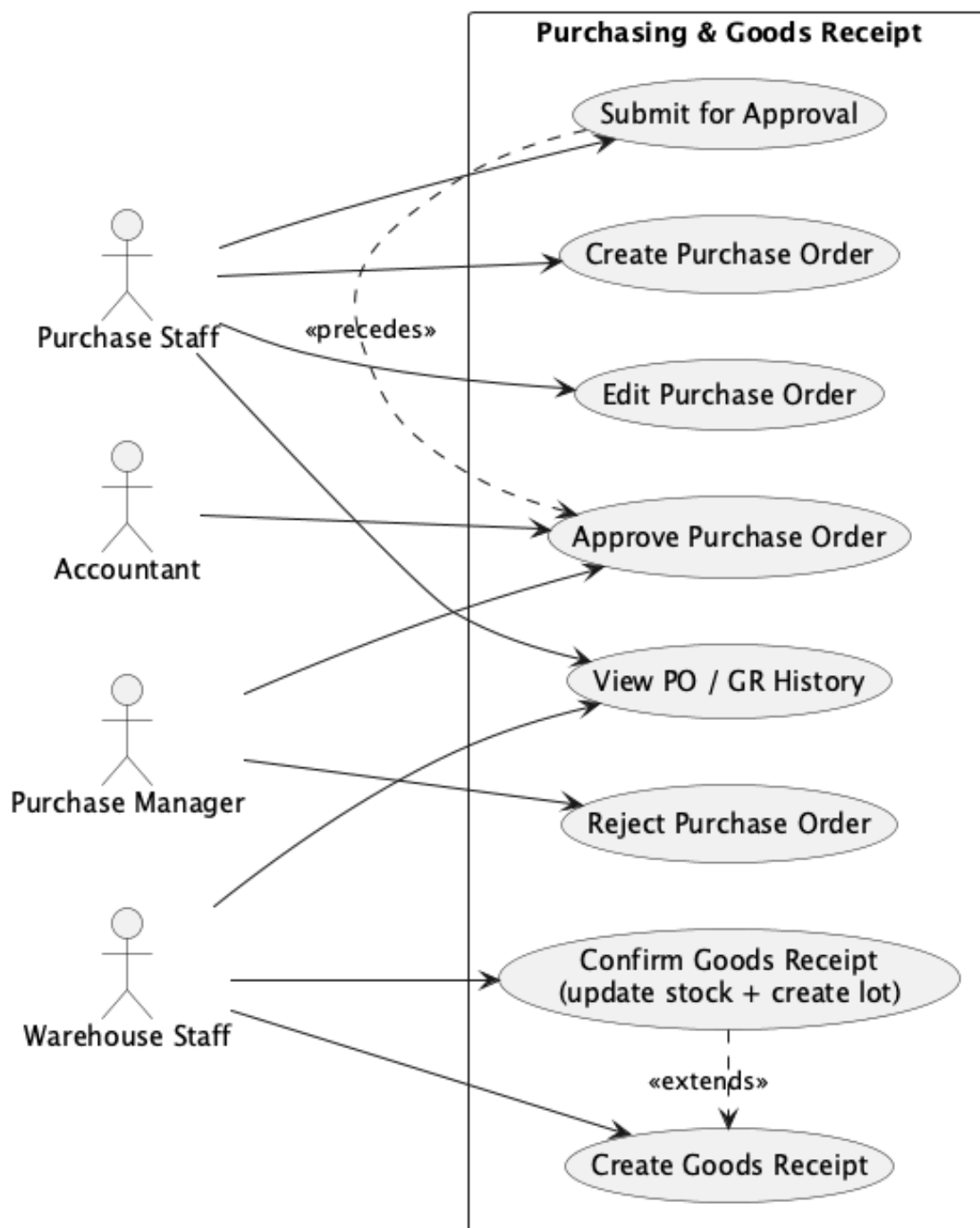
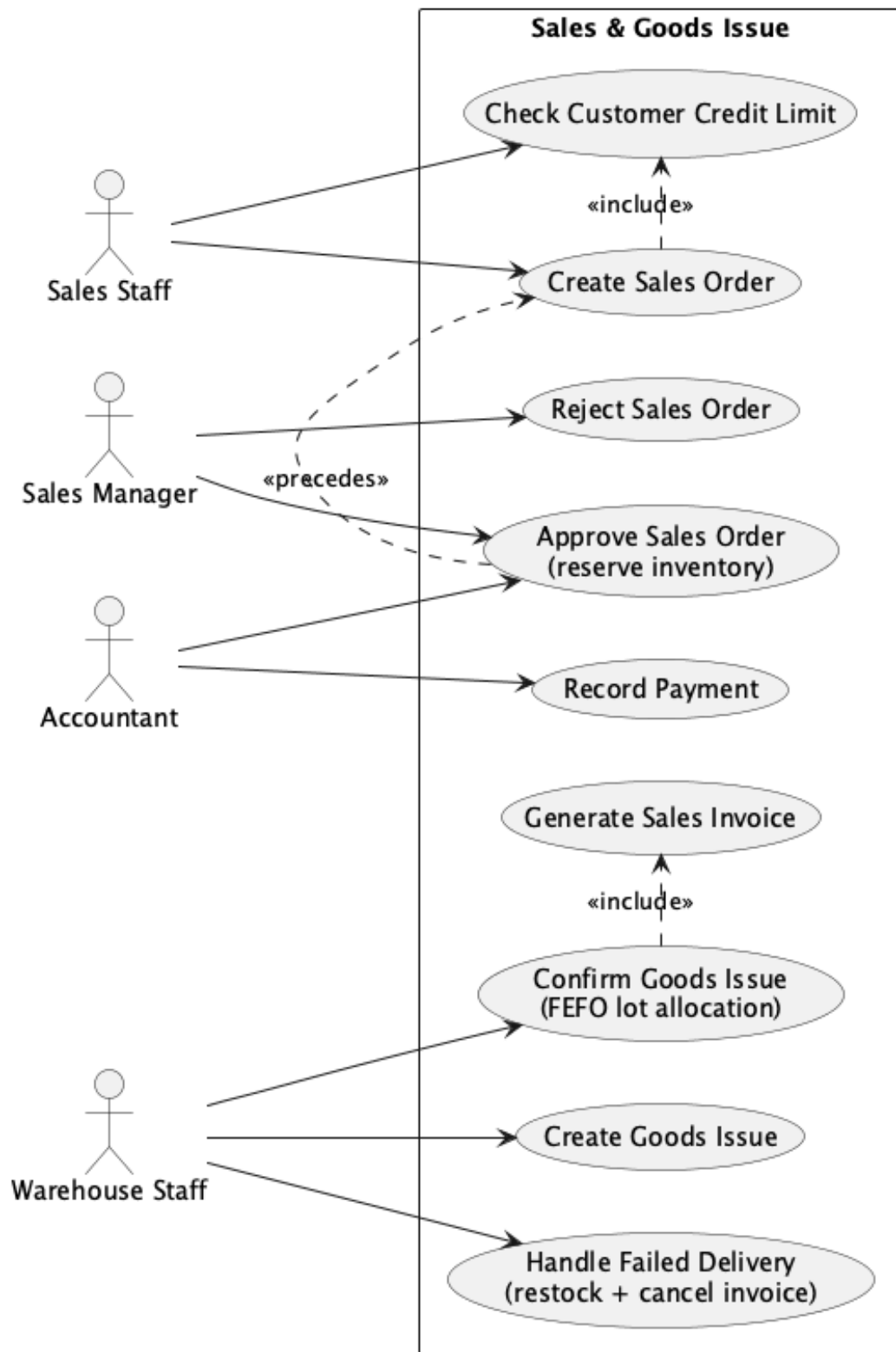


Figure 2.2: Detailed use cases — Purchasing & Goods Receipt

**Figure 2.3:** Detailed use cases — Sales & Goods Issue

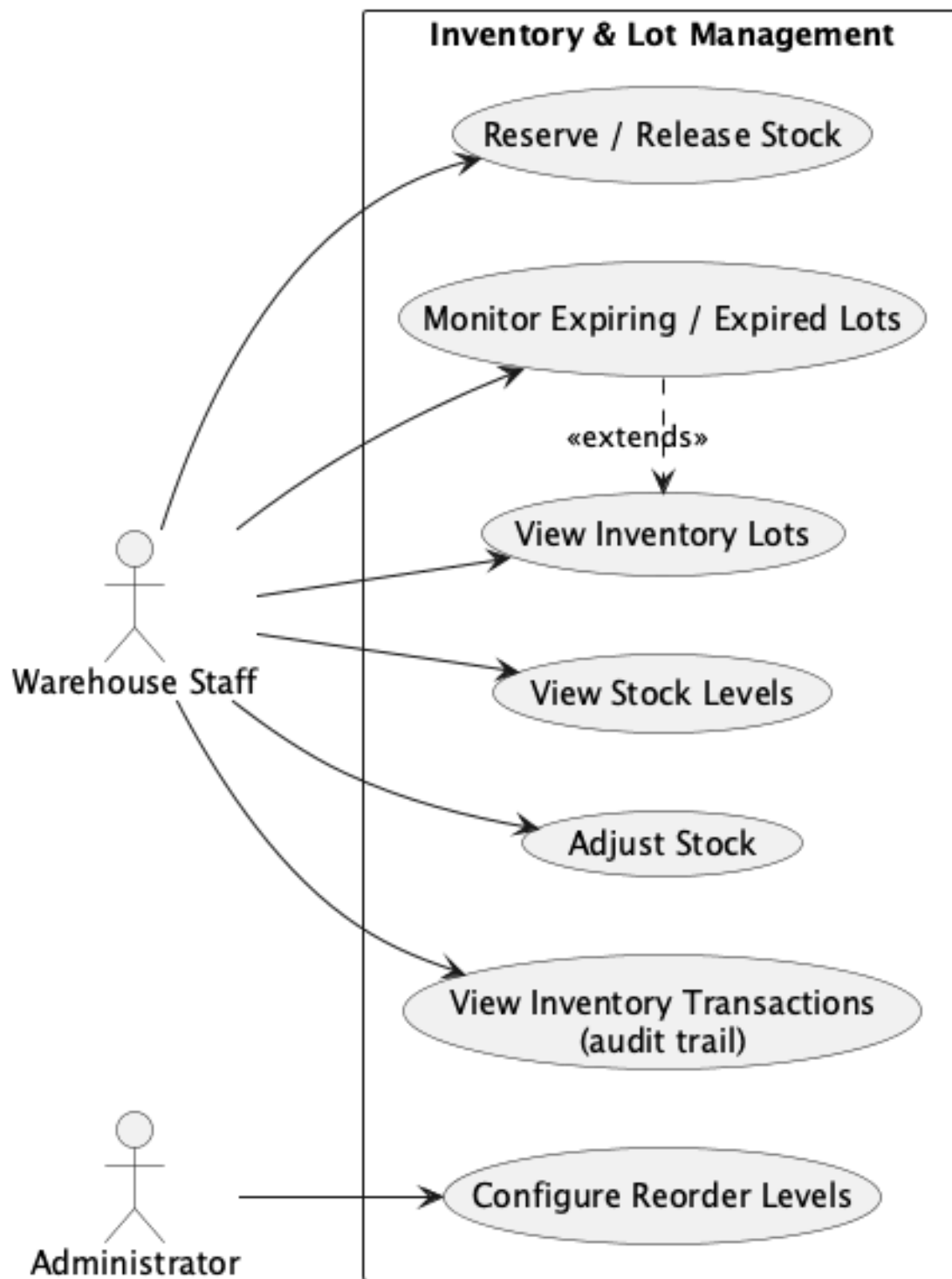


Figure 2.4: Detailed use cases — Inventory & Lot management

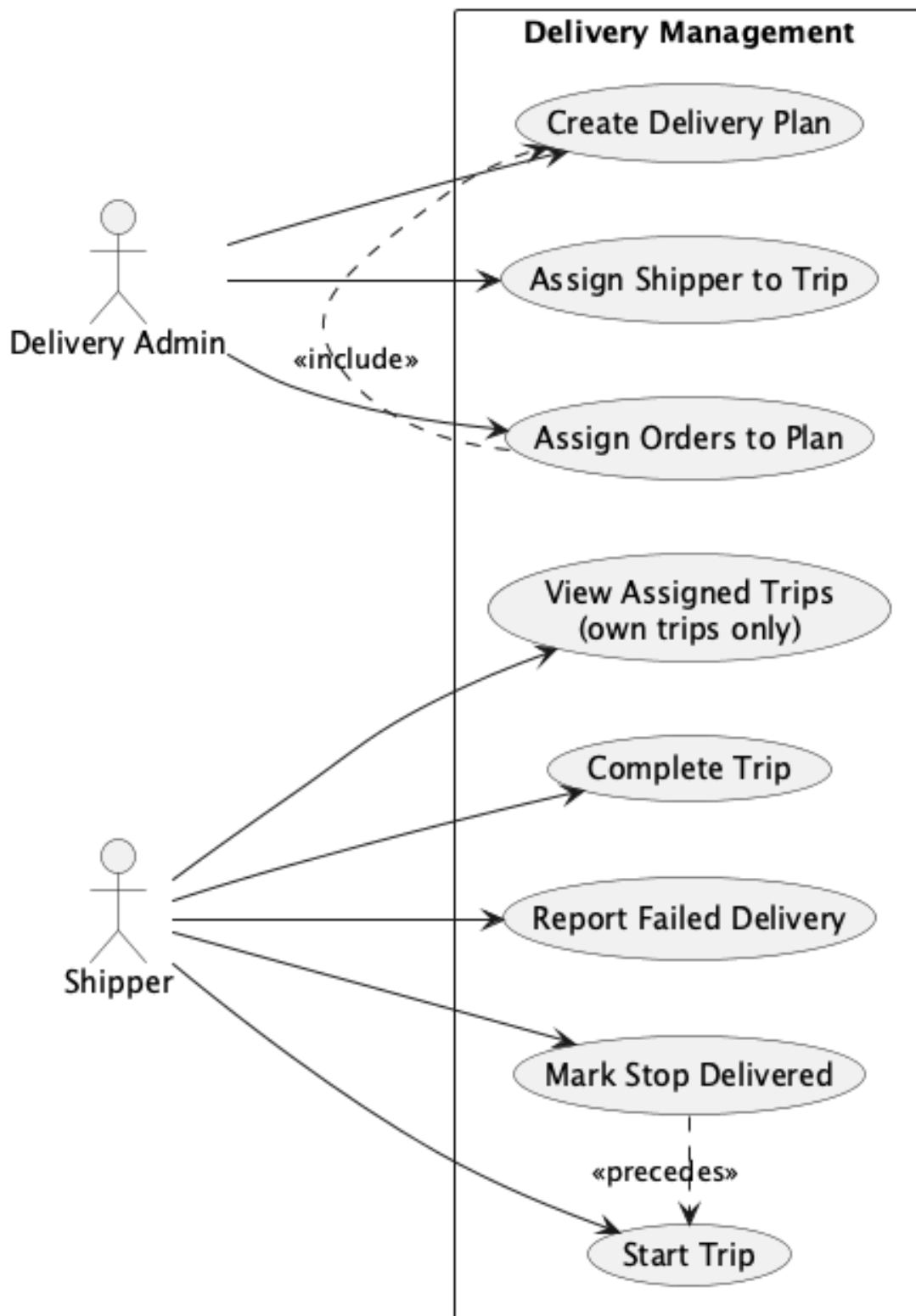


Figure 2.5: Detailed use cases — Delivery management

[TODO] Briefly explain each decomposed use case.

2.2.3 Business Process

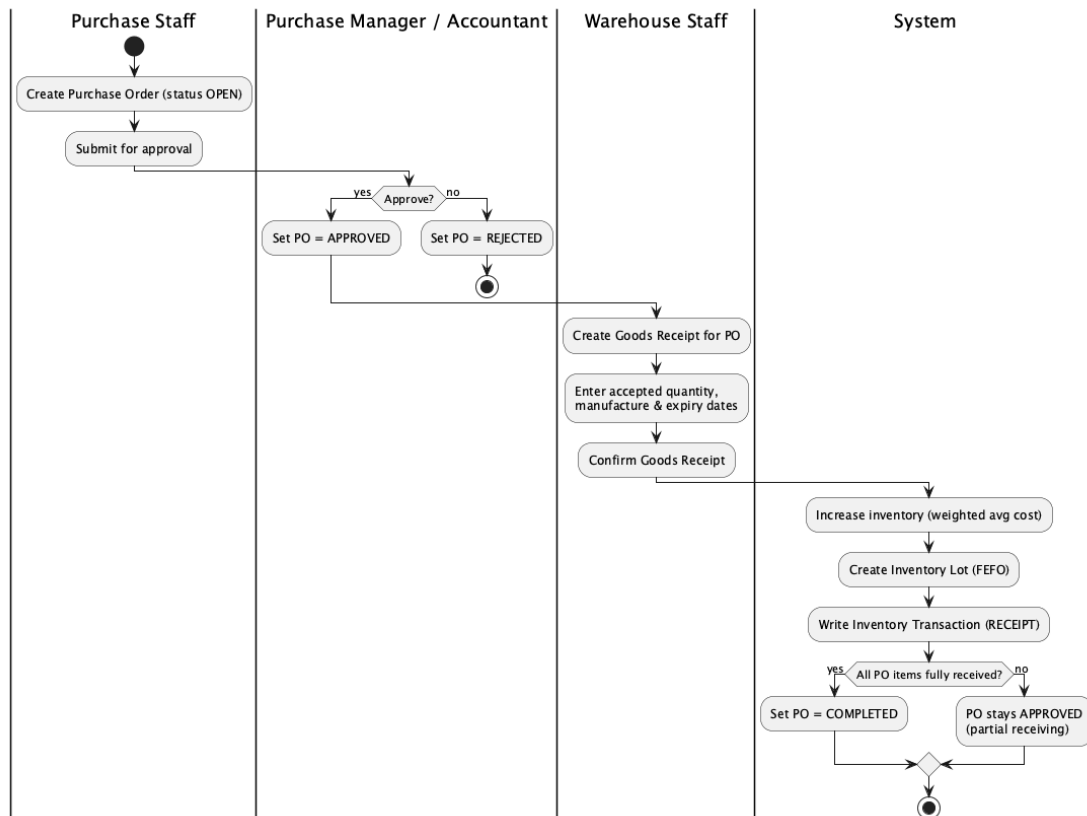


Figure 2.6: Procure-to-stock business process

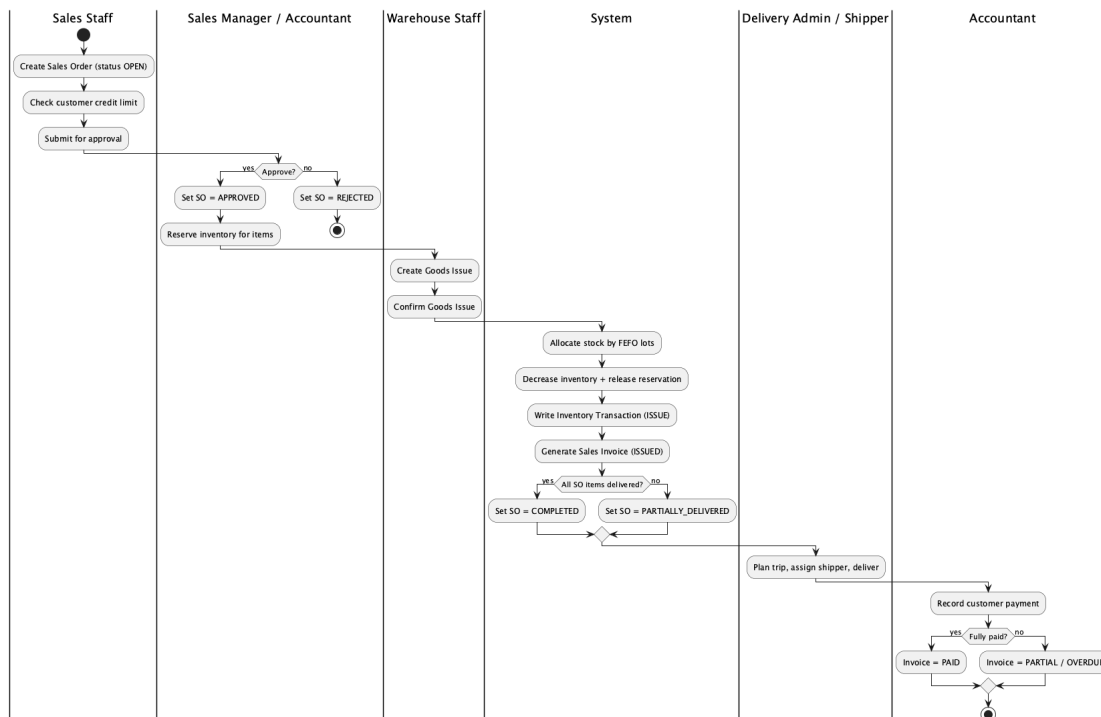


Figure 2.7: Order-to-cash business process

[TODO] Describe the two core business processes.

2.3 Functional Description

2.3.1 Confirm Goods Issue (FEFO)

- **Pre-conditions:** Sales Order is APPROVED; on-hand stock $\geq$ issued qty.
- **Main flow:** validate stock $\rightarrow$ allocate by FEFO lots $\rightarrow$ decrease inventory and release reservation $\rightarrow$ update delivered quantity $\rightarrow$ auto-generate invoice $\rightarrow$ update SO status.
- **Alternate flow:** insufficient lot stock $\rightarrow$ reject with error.
- **Post-conditions:** inventory and lots reduced; invoice issued.

2.3.2 Approve Sales Order (Inventory Reservation)

[TODO] Specify event flow, pre/post-conditions.

2.3.3 Confirm Goods Receipt

[TODO] Specify event flow, pre/post-conditions.

2.3.4 Assign and Execute Delivery Trip

[TODO] Specify event flow, pre/post-conditions.

2.4 Non-Functional Requirements

[TODO] List non-functional and technical requirements.

CHAPTER 3. THEORETICAL BACKGROUND AND TECHNOLOGIES

This chapter introduces the theoretical foundations and technologies used in the system. For each technology, the requirement it addresses, the available alternatives, and the rationale for the selection are discussed.

3.1 Inventory Management Theory

[TODO] Explain inventory concepts and FEFO; cite references ...

3.2 Backend Technologies

[TODO] Present the backend stack with justification.

3.3 Frontend Technologies

[TODO] Present the frontend stack with justification.

3.4 Database — PostgreSQL

[TODO] Justify the database choice.

3.5 Authentication and Authorization

[TODO] Explain JWT + RBAC theory as used in the system.

CHAPTER 4. DESIGN, IMPLEMENTATION, AND EVALUATION

This chapter presents the architecture, detailed design, implementation, testing, and deployment of the system.

4.1 Architecture Design

4.1.1 Software Architecture Selection

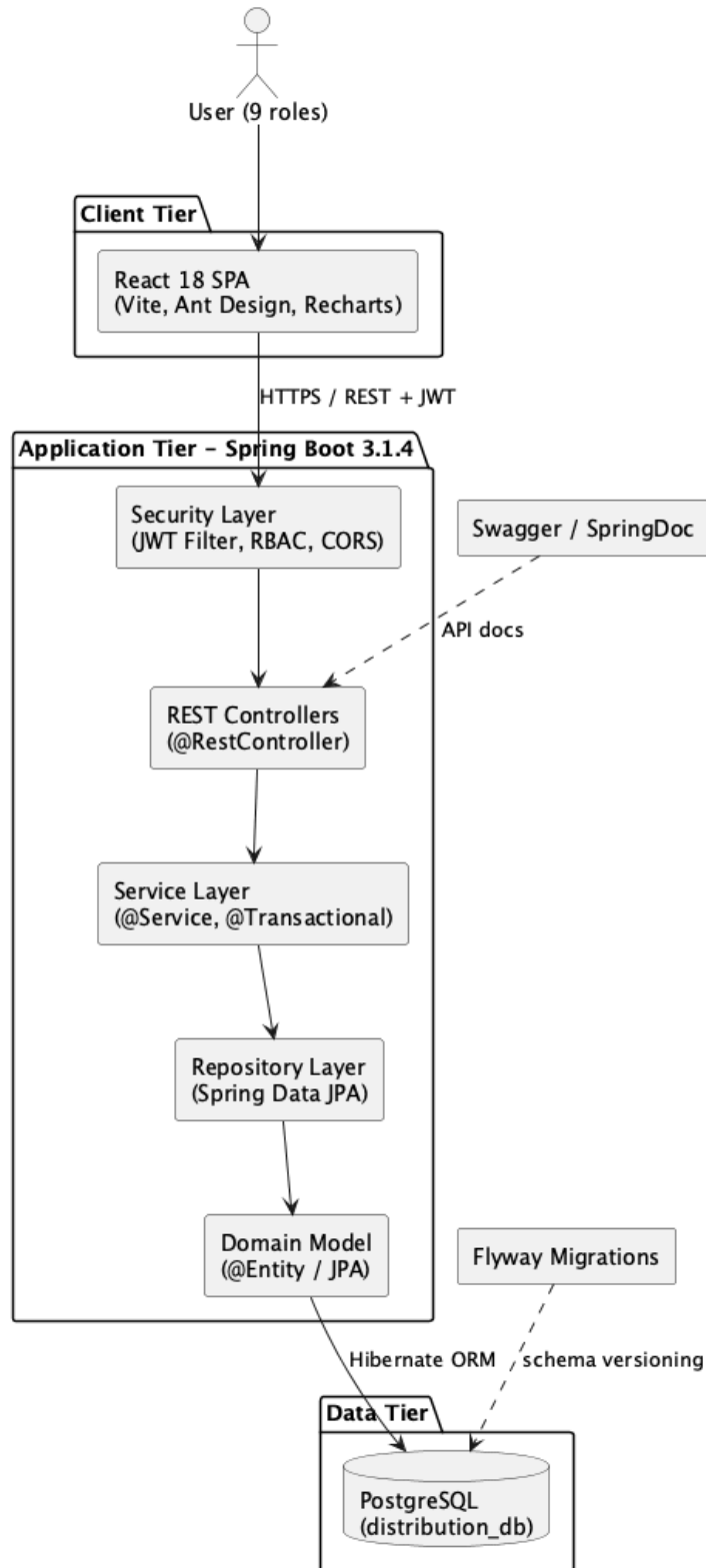


Figure 4.1: Overall system architecture

[TODO] Justify the layered/3-tier architecture and map it to the system.

4.1.2 Overall Design

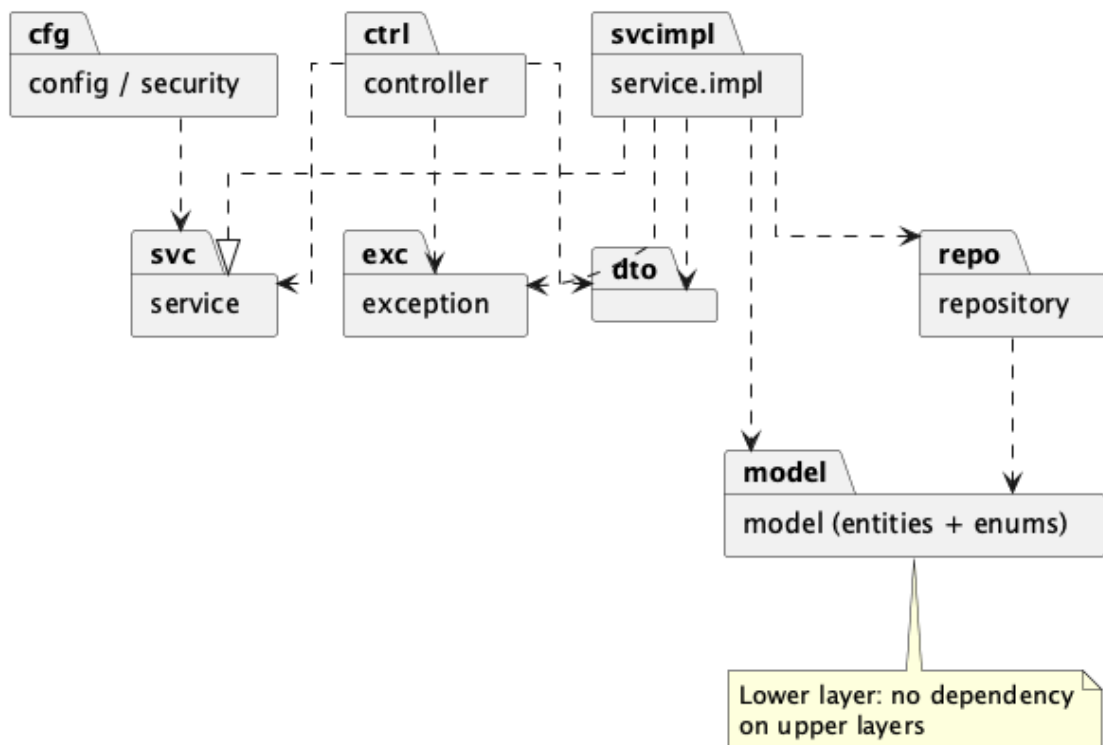


Figure 4.2: Package diagram (layered backend)

[TODO] Explain each package and the dependency direction.

4.1.3 Detailed Package Design

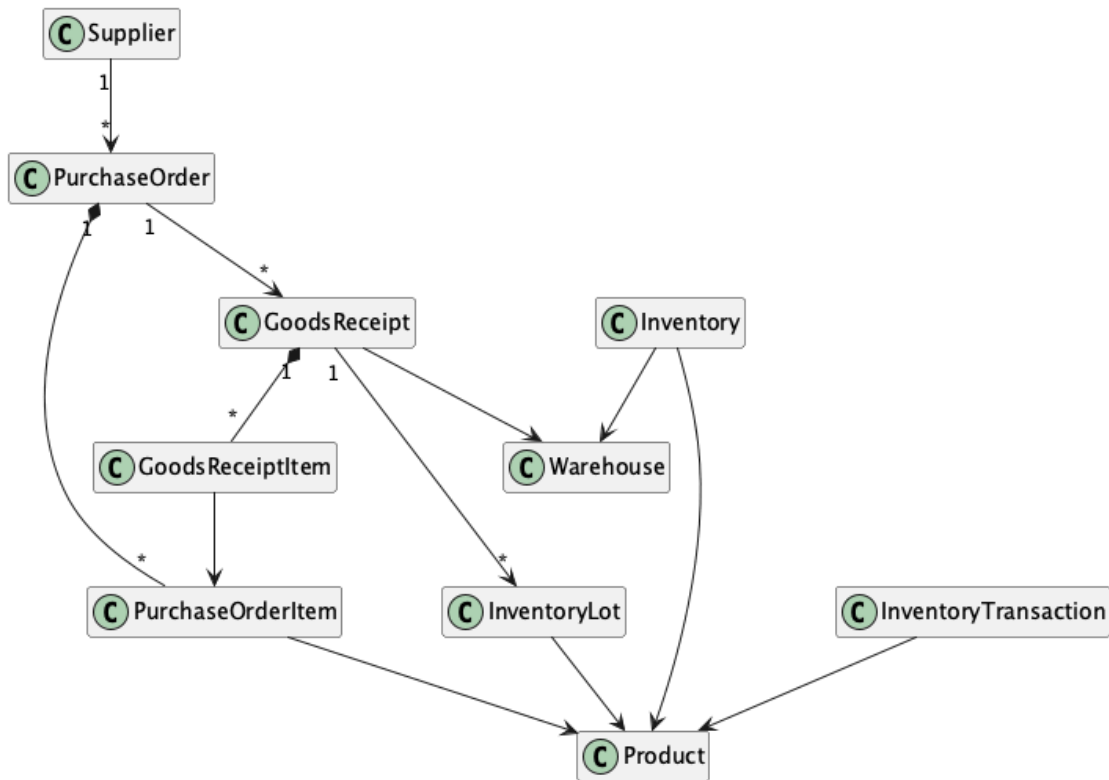


Figure 4.3: Class diagram — inbound (purchasing) flow

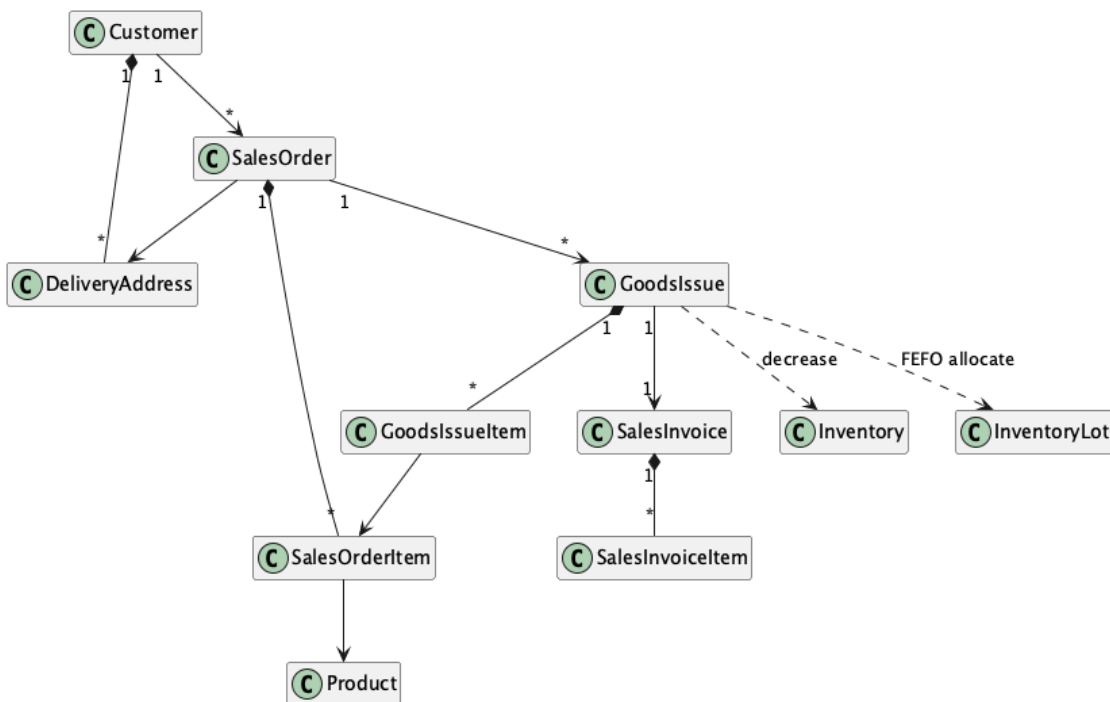


Figure 4.4: Class diagram — outbound (sales) flow

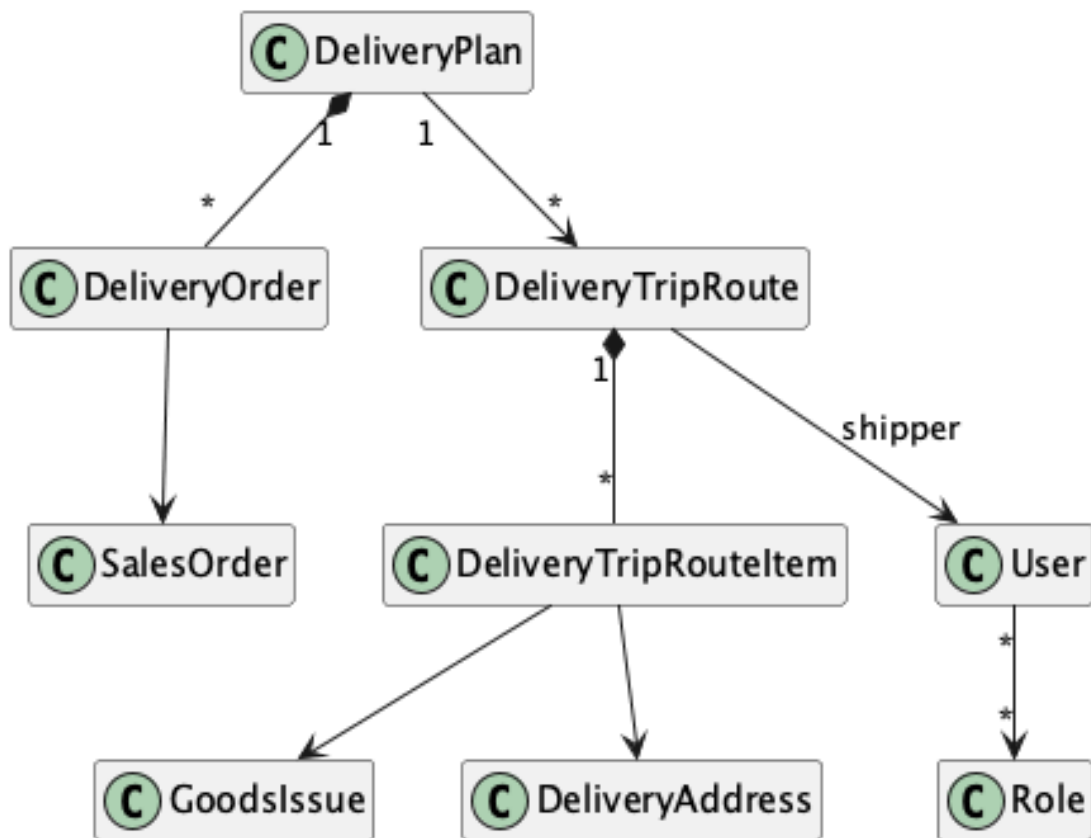


Figure 4.5: Class diagram — delivery flow

[TODO] Briefly explain the design of each package group.

4.2 Detailed Design

4.2.1 User Interface Design

[TODO] Present UI design standards and mockups (insert mockup figures here).

4.2.2 Layer Design

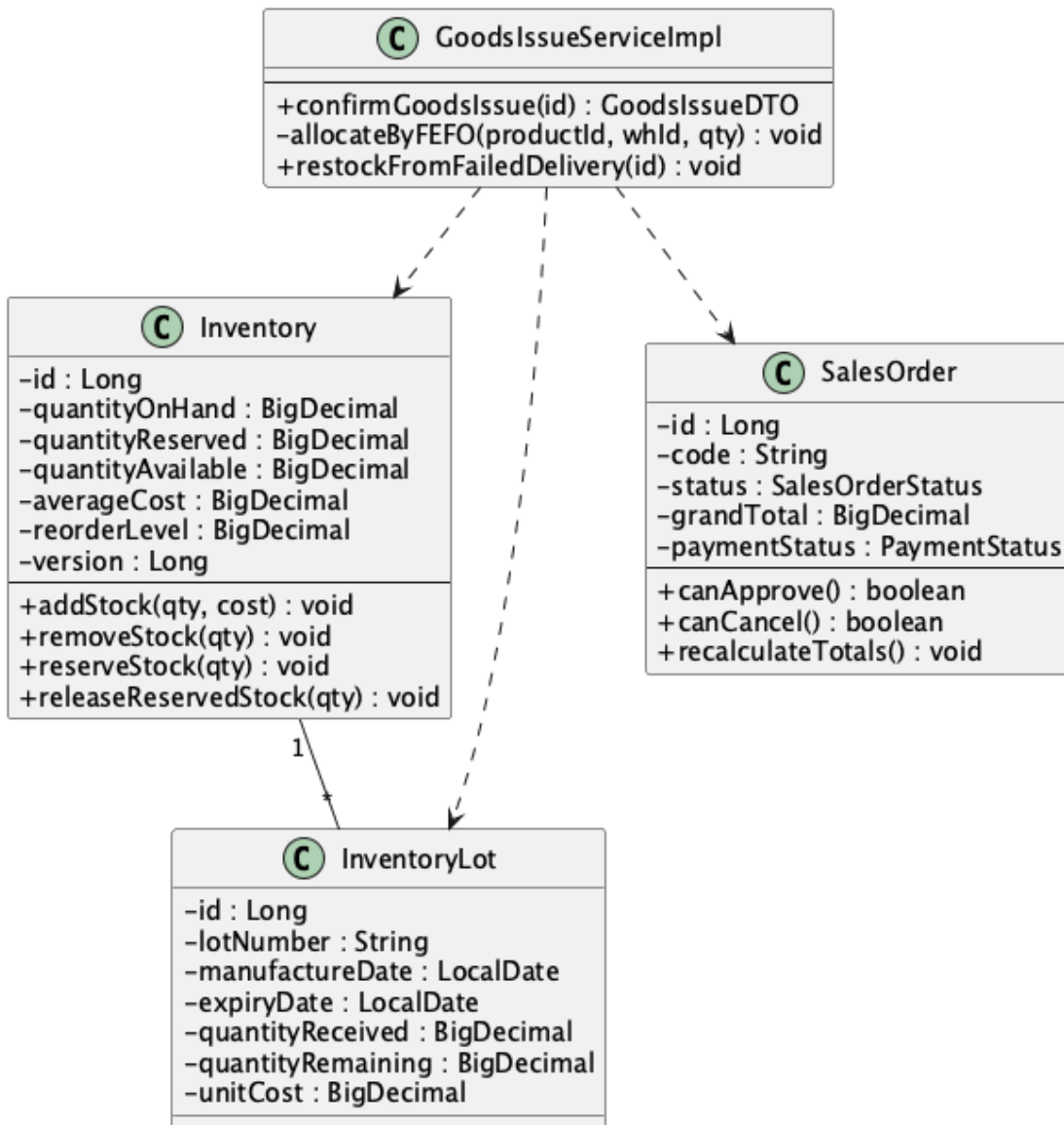


Figure 4.6: Detailed design of key classes

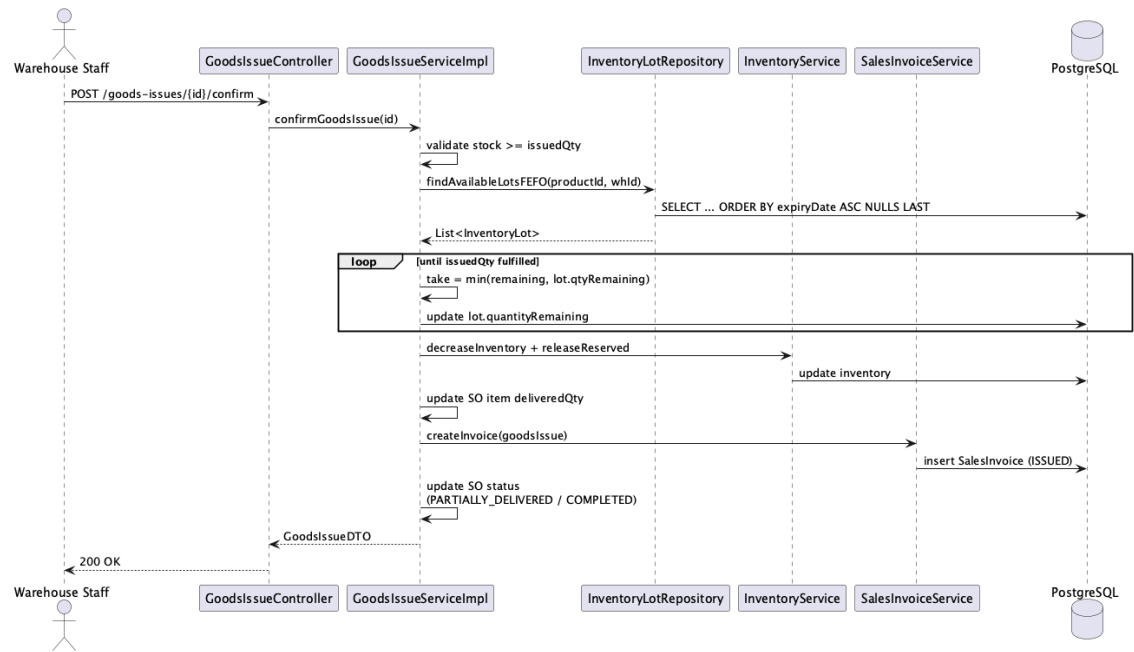


Figure 4.7: Sequence — confirm Goods Issue with FEFO allocation

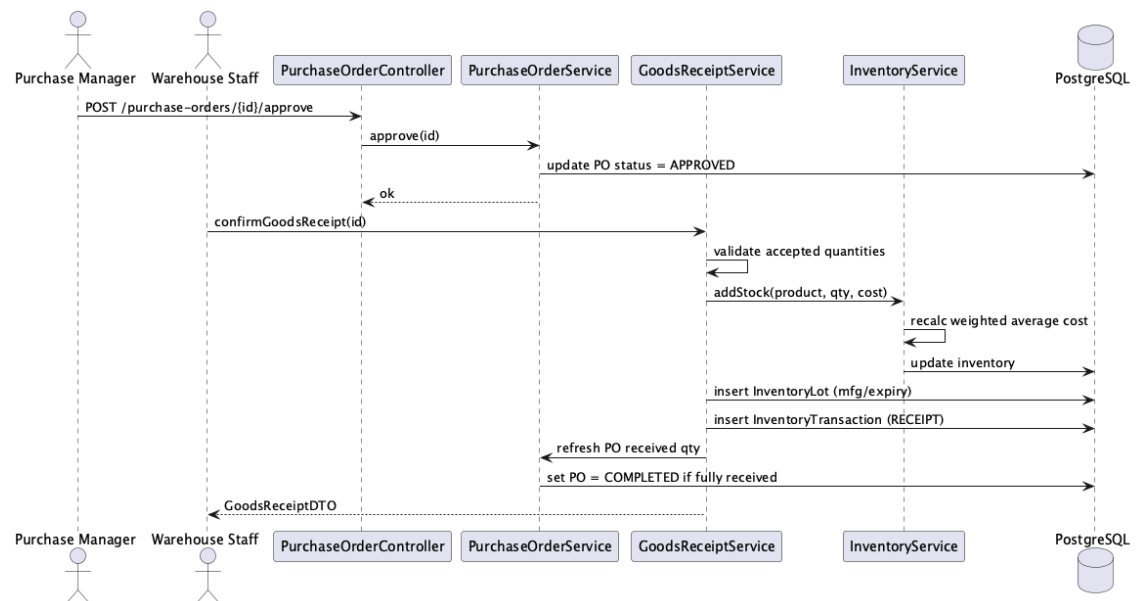


Figure 4.8: Sequence — approve PO and confirm Goods Receipt

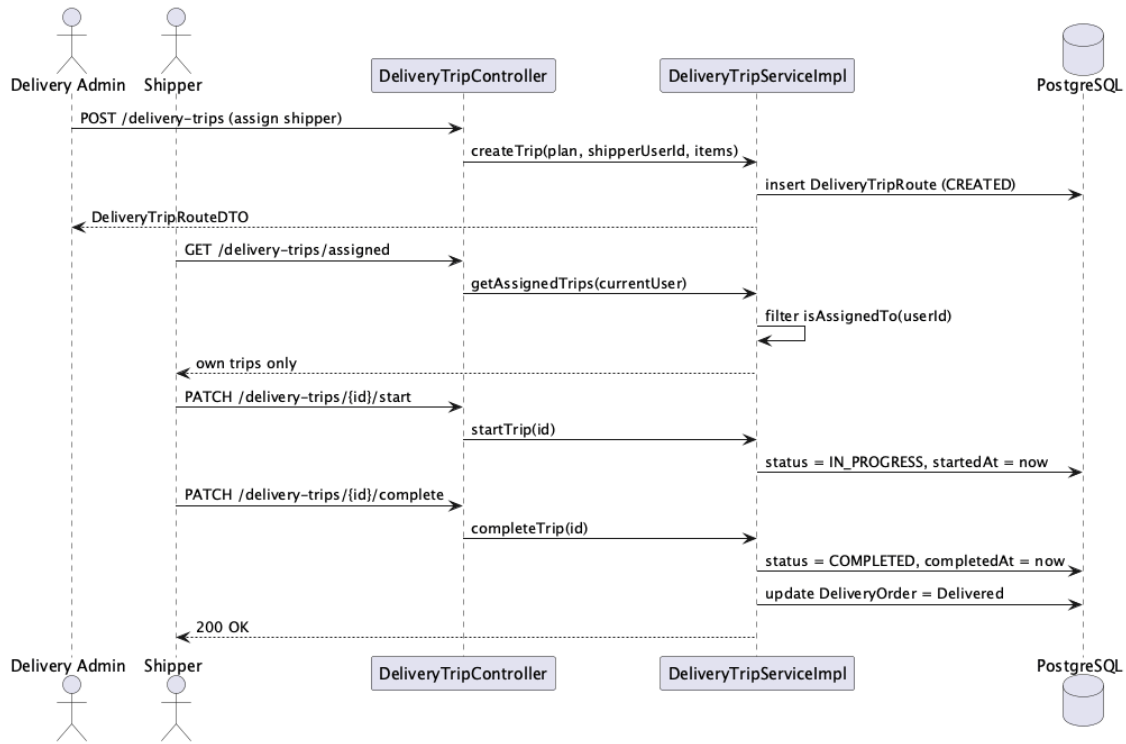


Figure 4.9: Sequence — assign and execute delivery trip

[TODO] Explain the key classes and the message flows.

4.2.3 Database Design

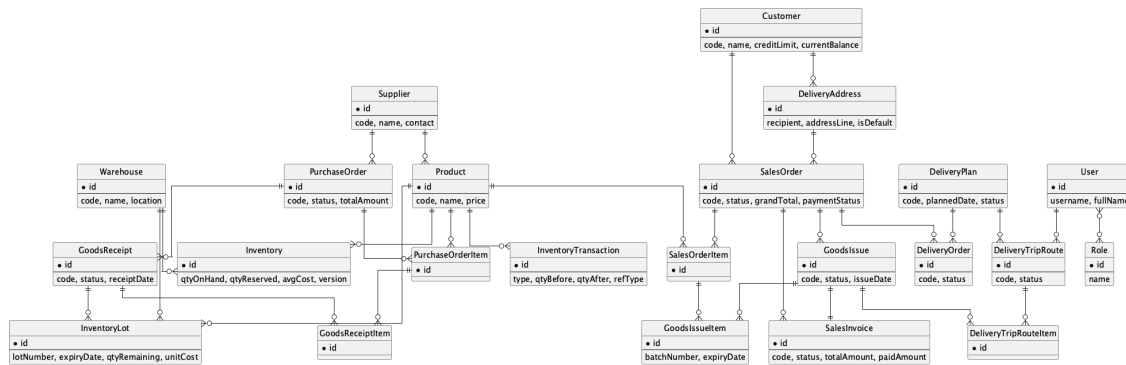


Figure 4.10: Entity-Relationship diagram

[TODO] Explain the entities, keys and relationships.

4.3 Application Building

4.3.1 Libraries and Tools

Table 4.1: Libraries, frameworks and tools used

Purpose	Tool / Library	Version
Backend language	Java	17
Backend framework	Spring Boot	3.1.4
ORM / data access	Spring Data JPA (Hibernate)	(Boot 3.1.4)
Security / tokens	Spring Security, JWT	0.12.3
DB migration	Flyway	(Boot-managed)
API documentation	SpringDoc OpenAPI	2.1.0
Boilerplate reduction	Lombok	(Boot-managed)
Build tool (backend)	Maven	3.x
Database	PostgreSQL	14+
Frontend library	React	18.2.0
Frontend build tool	Vite	5.0.0
UI components	Ant Design (antd)	5.10.0
HTTP client	Axios	1.4.0
Routing	React Router DOM	6.11.1
Charts	Recharts	3.8.1
Date/time	Day.js	1.11.10
IDE	IntelliJ IDEA / VS Code	–

4.3.2 Achievement

Table 4.2: Source-code statistics

Metric	Value
Lines of code (backend)	[TODO from cloc]
Lines of code (frontend)	[TODO from cloc]
Number of Java classes	[TODO]
Number of packages	[TODO]

4.3.3 Illustration of Main Functions

[TODO] Insert and explain annotated screenshots of the main functions.

4.4 Testing

Table 4.3: Sample test cases

ID	Function	Input	Expected	Result
TC01	Confirm GI (FEFO)	stock with 2 lots	earliest-expiry lot consumed first	[TODO]
TC02	Approve SO	qty ≤ available	reservation increases	[TODO]
TC03	Credit limit	balance+amt > limit	order blocked	[TODO]

[TODO] Describe testing technique and summarise results.

4.5 Deployment

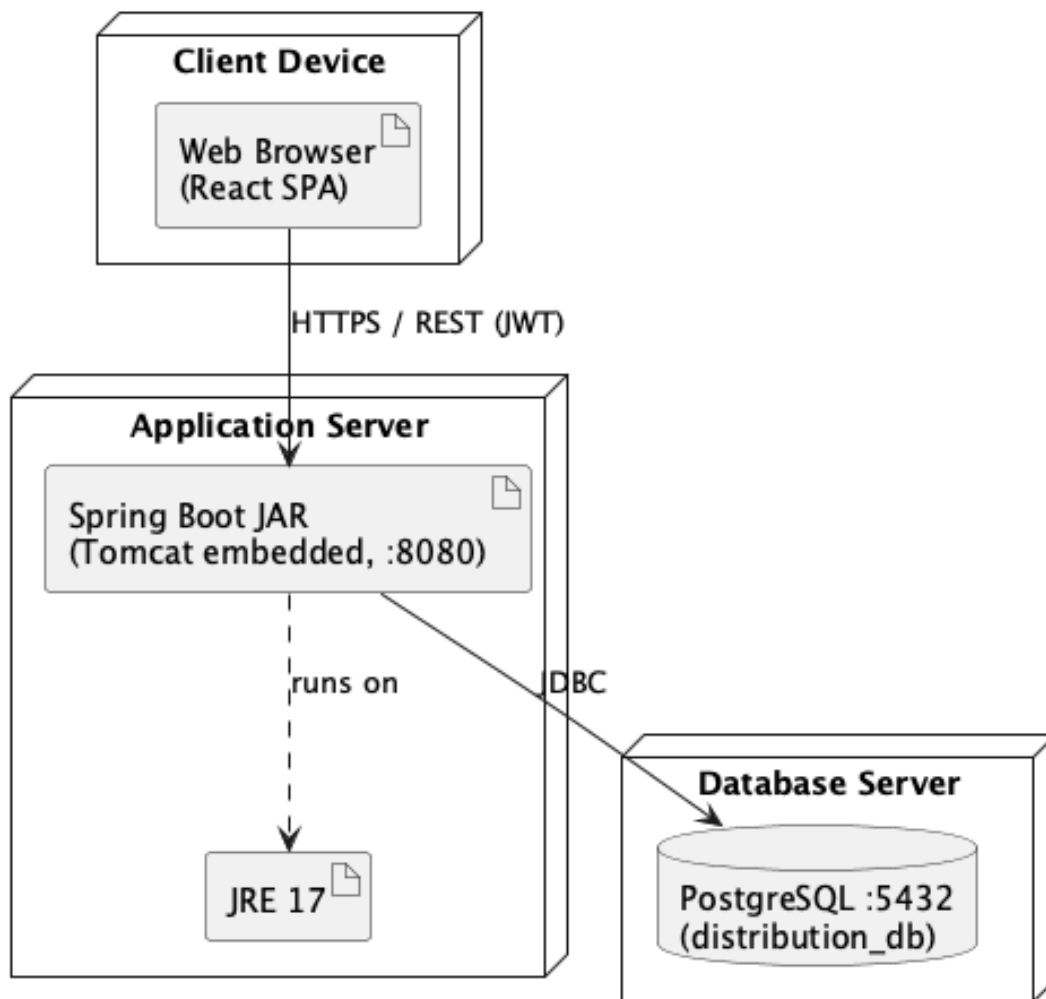


Figure 4.11: Deployment diagram

[TODO] Describe the deployment model and environment.

CHAPTER 5. SOLUTION AND CONTRIBUTION

This chapter presents the principal contributions of the thesis. Each contribution is described through the problem it addresses, the proposed solution, and the results achieved.

5.1 FEFO Lot-Allocation Mechanism

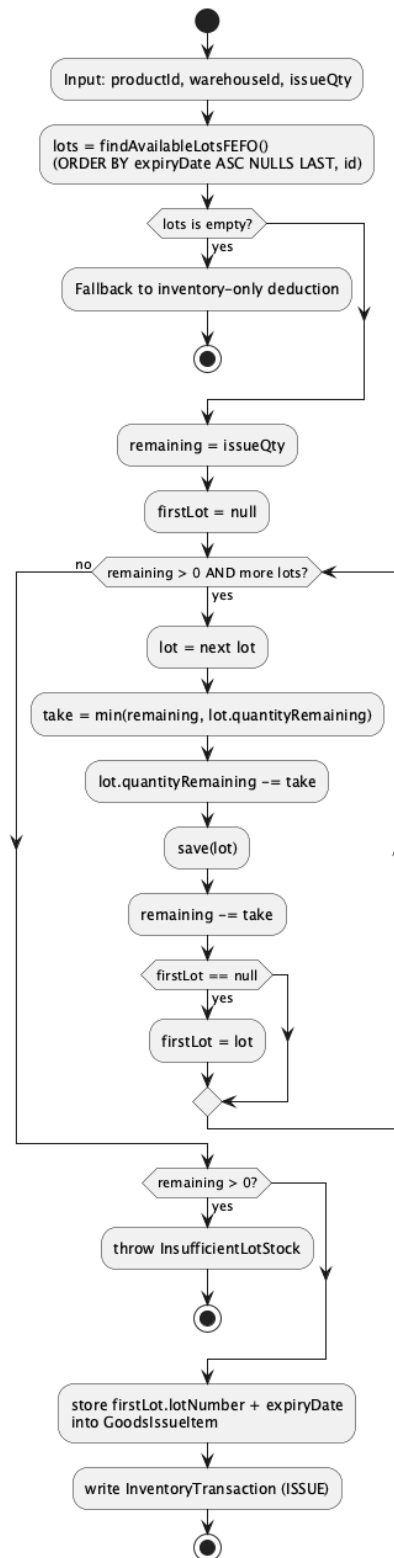


Figure 5.1: FEFO allocation algorithm

[TODO] Present the problem, the FEFO algorithm, and the results.

5.2 Inventory Reservation on Order Approval

[TODO] Present the reservation design and its effect.

5.3 Failed-Delivery Recovery

[TODO] Present the recovery mechanism.

5.4 Role-Based Workflow Enforcement

[TODO] Present the RBAC design; optionally include a permission matrix table.

CHAPTER 6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

[TODO] Summarise achievements, comparison and lessons learned.

6.2 Future Work

[TODO] Describe planned completion work and new directions.

APPENDIX

A. ADDITIONAL USE CASE DESCRIPTIONS

Use case: Record Customer Payment

[TODO] Specify the use case.

Use case: Adjust Stock

[TODO] Specify the use case.

Use case: Monitor Expiring Lots

[TODO] Specify the use case.